

Number of Functional Features Included in the Solution

DATE	02 JULY 2026
TEAM ID	
PROJECT NAME	A Comprehensive Measure of Well-Being: Human Development Index (HDI) Predictor
MAXIMUM MARKS	5 MARKS

1. Overview

The HDI Prediction System incorporates six primary functional features to provide an end-to-end, production-grade tool for simulating and analyzing human development levels.

2. Core Functional Features

2.1 Feature 1: Multi-Indicator Form with Input Validation

- **Description:** An interactive input form that allows users to enter values for the target country and the four developmental indicators.
- **Implementation:** Form fields use native HTML5 validation constraints to prevent out-of-range inputs (e.g. Life Expectancy must be between 50 and 85 years).
- **Benefit:** Prevents incorrect inputs and ensures prediction stability.

2.2 Feature 2: Machine Learning Prediction Pipeline

- **Description:** A predictive model that uses Linear Regression to calculate a continuous HDI score based on the user's inputs.
- **Implementation:** The backend loads ``models/hdi_model.pkl`` and ``models/encoder.pkl``, encodes the country, and generates the prediction with low latency (<200ms).
- **Benefit:** Provides instant predictions based on user inputs.

2.3 Feature 3: Standard UNDP Tier Classification

- **Description:** Automatically classifies the continuous predicted score into one of the four standard development tiers defined by the United Nations: Very High (≥ 0.800), High (0.700 to 0.799), Medium (0.550 to 0.699), and Low (< 0.550).
- **Implementation:** Executed on the backend using boundary checks in ``classify_hdi(score)``.
- **Benefit:** Groups countries into clear development categories.

2.4 Feature 4: Contextual Recommendation Engine

- **Description:** Generates a list of targeted policy recommendations and development goals based on the predicted tier.
- **Implementation:** Managed by ``get_hdi_insights(tier)`` in ``app.py``, which returns recommendations tailored to the predicted level.
- **Benefit:** Provides policymakers with actionable next steps to improve development outcomes.

2.5 Feature 5: Visual Gradient Progress Gauge

- **Description:** A visual gauge bar that indicates the relative position of the predicted score on the human development spectrum.
- **Implementation:** Built using dynamic CSS variables that map the score directly to the fill width of a color-gradient progress bar.
- **Benefit:** Provides an intuitive, visual representation of the predicted score.

2.6 Feature 6: Interactive Educational Dashboard

- **Description:** A landing page explaining the three dimensions of HDI (Health, Education, and Income) to provide context for users.
- **Implementation:** Designed using responsive glassmorphism cards and clean layout templates.
- **Benefit:** Helps users understand the factors that drive human development.